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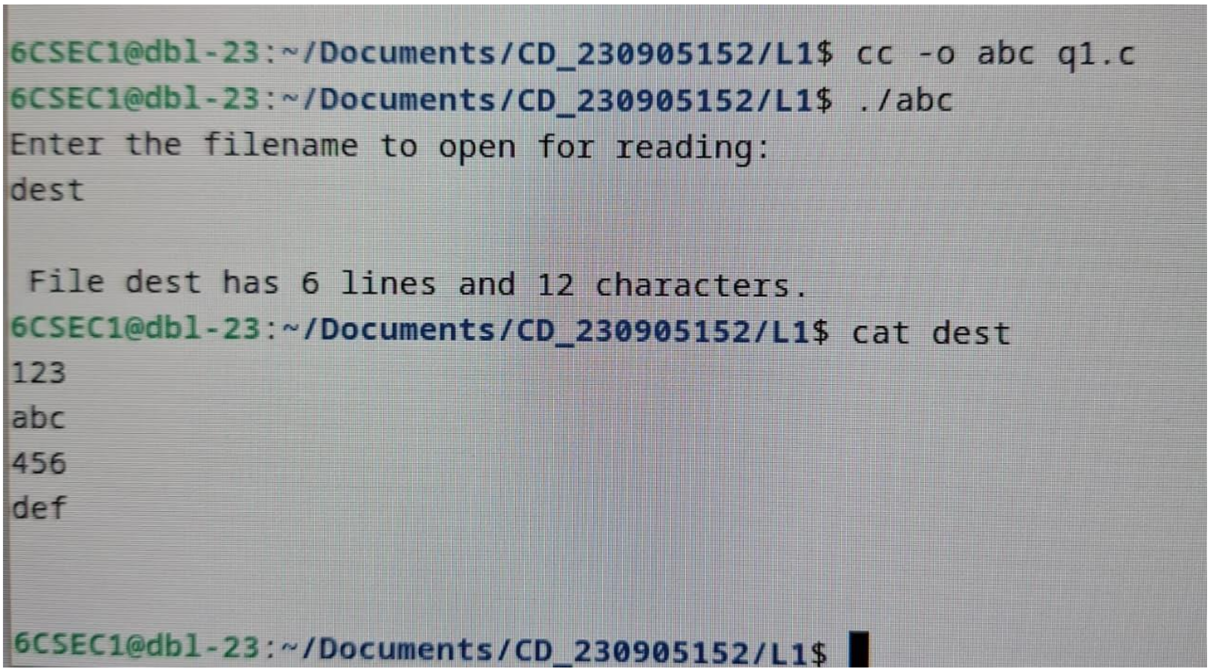
Q1.

```
#include <stdio.h>

#include <stdlib.h>

int main(){

	FILE *fptr1;

	char filename[100], c;

	printf("Enter the filename to open for reading: \n");

	scanf("%s", filename);

	fptr1 = fopen(filename, "r");

	if (fptr1 == NULL){

		printf("Cannot open file %s \n", filename);

		exit(0);

	}

	int ct_lines=0;

	int ct_chars=0;

	c = fgetc(fptr1); // Read contents from file

	while (c != EOF){

		if(c=='\n') ++ct_lines;

		else ++ct_chars;

		c = fgetc(fptr1);

	}

	printf("\n File %s has %d lines and %d characters.\n", filename,ct_lines,ct_chars);

	fclose(fptr1);

	return 0;

}
```

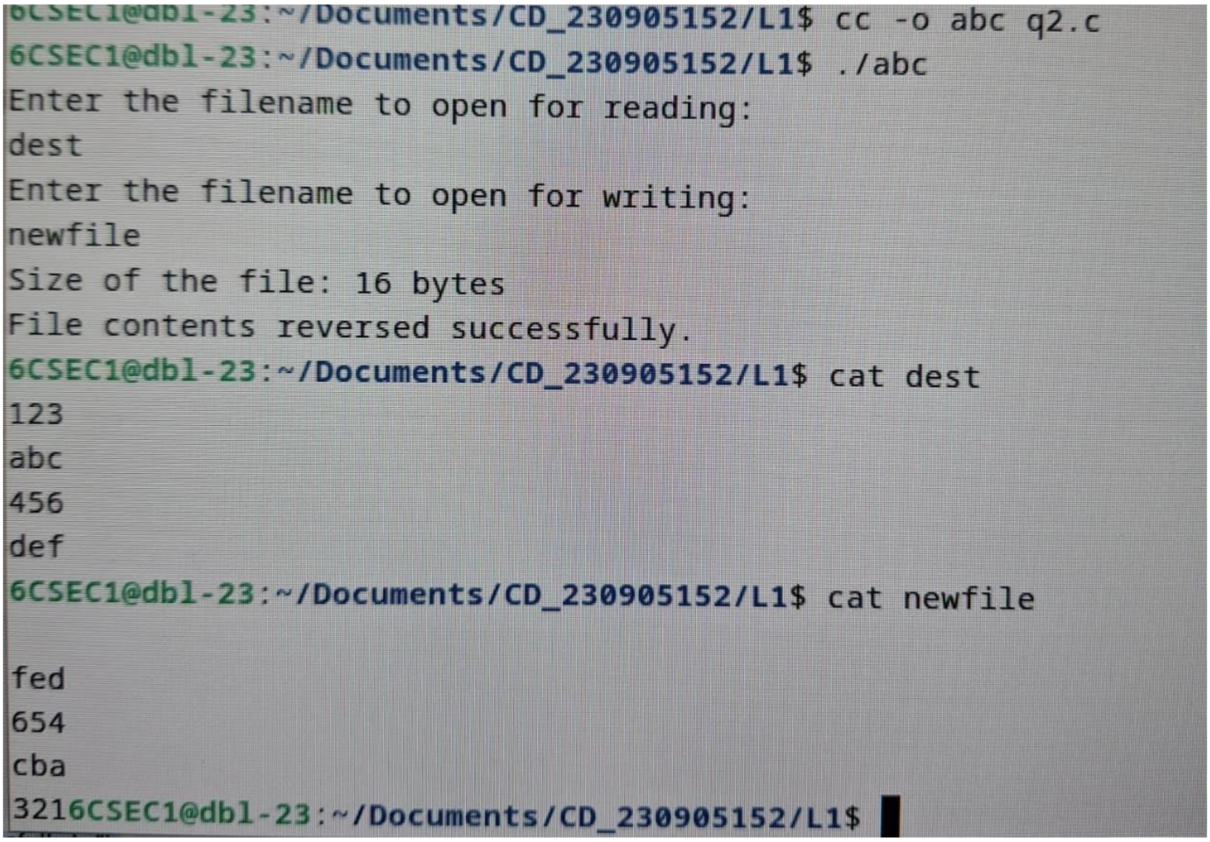


```
6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ cc -o abc q1.c
6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ ./abc
Enter the filename to open for reading:
dest

 File dest has 6 lines and 12 characters.
6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ cat dest
123
abc
456
def


6CSEC1@dbl-23:~/Documents/CD_230905152/L1$
```

Q2.

```
#include <stdio.h>

#include <stdlib.h>

int main() {

        FILE *fp1, *fp2; long size; char filename[100],ch;

        printf("Enter the filename to open for reading: \n");

        scanf("%s", filename);

        fp1 = fopen(filename, "r");

        printf("Enter the filename to open for writing: \n");

        scanf("%s", filename);

        fp2 = fopen(filename, "w+");

        fseek(fp1, 0, SEEK_END);

        size = ftell(fp1);

        printf("Size of the file: %ld bytes\n", size);

        for (long i = 1; i <= size; i++) {

                fseek(fp1, -i, SEEK_END);

                ch = fgetc(fp1);

                  fputc(ch, fp2);

        }
```

```
	fclose(fp1);

	fclose(fp2);

	printf("File contents reversed successfully.\n");

	return 0;

}
```

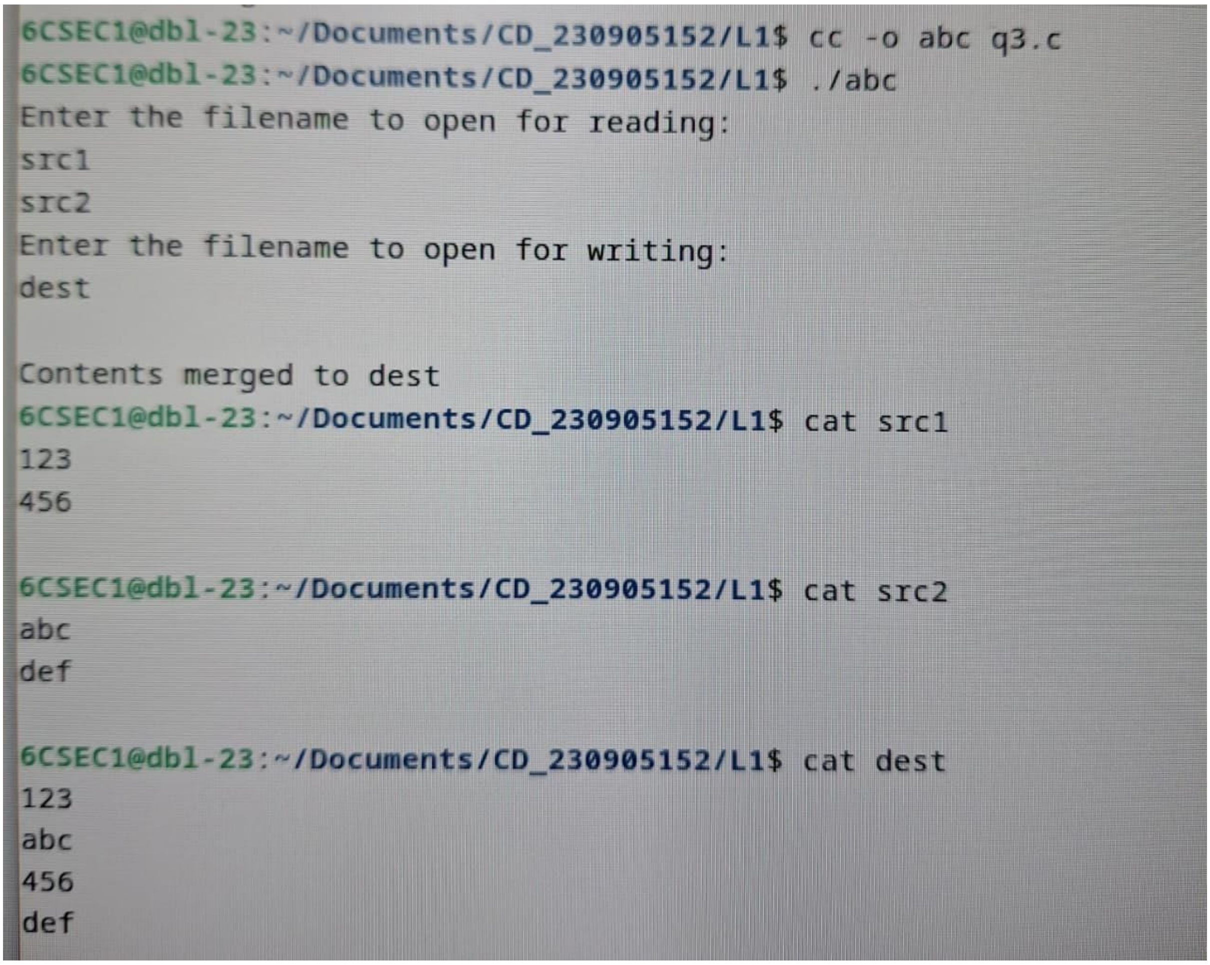


Q3.

```
#include <stdio.h>

#include <stdlib.h>

int main(){

	FILE *fptr1, *fptr2, *fptr3;

	char filename[100], c;

	printf("Enter the filename to open for reading: \n");

	scanf("%s", filename);

	fptr1 = fopen(filename, "r");

	if (fptr1 == NULL){

		printf("Cannot open file %s \n", filename);

		exit(0);
```

```
}
scanf("%s", filename);
fptr2 = fopen(filename, "r");
if (fptr2 == NULL){
        printf("Cannot open file %s \n", filename);
        exit(0);
}
printf("Enter the filename to open for writing: \n");
scanf("%s", filename);
fptr3 = fopen(filename, "w+");
int f1_done = 0, f2_done = 0;
while (!f1_done || !f2_done) {
  if (!f1_done) {
        while (1) {
          c = fgetc(fptr1);
          if (c == EOF) {
            f1_done = 1;
            break;
          }
          fputc(c, fptr3);
          if (c == '\n') break;
        }
  }
  if (!f2_done) {
        while (1) {
          c = fgetc(fptr2);
          if (c == EOF) {
            f2_done = 1;
            break;
          }
          fputc(c, fptr3);
```

```
                if (c == '\n') break;
            }
    }
    }
    printf("\nContents merged to %s \n", filename);
    fclose(fptr1);
    fclose(fptr2);
    fclose(fptr3);
    return 0;
}
```



```
6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ cc -o abc q3.c
6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ ./abc
Enter the filename to open for reading:
src1
src2
Enter the filename to open for writing:
dest

Contents merged to dest
6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ cat src1
123
456

6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ cat src2
abc
def

6CSEC1@dbl-23:~/Documents/CD_230905152/L1$ cat dest
123
abc
456
def
```